Technical Report: Cleaning, Visualization, and Exploratory Analysis of the Birds of the World Dataset

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Dataset source: Birds of the World (Unprocessed), Kaggle
https://www.kaggle.com/datasets/anoopjohny/birdsoftheworld-unprocessed

Abstract

This report presents the cleaning, reconstruction, visualization, and exploratory analysis of unprocessed the Birds of the World dataset. The raw CSV file contained malformed rows caused by unquoted commas inside text fields, which prevented reliable parsing through a standard CSV reader. To address this issue, a custom Python workflow was used to rebuild records by anchoring each row on the timestamp field and reconstructing the intended six logical columns: species, location, time, description, sex, and feather color. After reconstruction, the dataset was standardized, deduplicated, and enriched with helper features such as year, hour, description length, and tokenized color labels. Descriptive analysis shows broad species diversity, heavy concentration of records in a small number of locations, a highly imbalanced sex field dominated by Unknown entries, and repeated mention of colors such as white, black, brown, yellow, and red. The resulting dataset is suitable for exploratory analysis, although source-quality issues still limit deeper biological interpretation.

Keywords: Data Cleaning, Exploratory Data Analysis, Python, Pandas, Matplotlib, Bird Observations, CSV Reconstruction, Visualization

1 Introduction

Exploratory data analysis depends on reliable structure, yet many real-world datasets arrive in a form that is incomplete, inconsistent, or badly serialized. In this project, the Birds of the World dataset was selected as a case study in practical data cleaning because its raw file contains broken rows produced by commas embedded in text fields. As a result, a standard pandas.read_csv() import creates overflow columns and an unreliable table structure.

The purpose of this report is to document how the dataset was reconstructed into a usable analytical format, how the cleaned data was summarized through descriptive statistics and visualizations, and what major patterns were observed after cleaning. The report follows the same end-to-end workflow expected in an applied data science task: inspect, clean, transform, visualize, interpret, and document limitations.

2 Objectives

The project was guided by the following objectives:

- Reconstruct malformed CSV rows into the intended six-column structure.
- Standardize and clean the recovered dataset by parsing dates, trimming text, and removing exact duplicates.
- Create derived analytical features such as year, hour, description length, and feather color counts.
- Visualize species frequency, color patterns, temporal trends, and sex label distribution.
- Evaluate the analytical value and limitations of the dataset after reconstruction.

3 Methodology

3.1 Dataset Source and Structure

The analysis used the Birds of the World (Unprocessed) dataset obtained from Kaggle. The source file was intended to contain six core variables: species, location, time, description, sex, and feather color. However, commas inside free-text fields were not always quoted, causing rows to spill across extra columns when read with default CSV settings.

3.2 Cleaning Workflow

A custom reconstruction step was used to recover the intended schema. Each row was read with Python's csv module, then matched against a regular expression that identified the

timestamp pattern dd-mm-yyyy hh:mm. Values before the timestamp were reassembled into species and location fields, while values after the timestamp were rebuilt into description, sex, and feather color fields.

Once the raw structure was restored, the workflow standardized text fields, parsed timestamps into datetime values, removed exact duplicate observations, and created helper variables such as year, month, hour, description length, and tokenized feather-color labels. The cleaned dataset was then used for summary statistics and visualization in Pandas and Matplotlib.

4 Results and Analysis

4.1 Cleaning Outcome

The reconstruction pipeline recovered 1,310 rows with six meaningful columns. After removing 8 exact duplicate observations, 1,302 rows remained for descriptive analysis. The cleaned data contained 534 unique species, 124 unique locations, and a date range spanning from 2001 to 2023. These results indicate that the dataset became analytically usable after reconstruction, even though it originated from a malformed CSV file.

4.2 Interpretation of Major Patterns

Frequency analysis showed that observations are concentrated in a small number of common species, with *Ruby-throated Hummingbird, American Goldfinch, Blue Jay,* and *Northern Flicker* appearing among the most frequent entries. Location counts also suggest that observations are clustered within broad labels such as the United States, which limits precision in geographic interpretation.

Feather-color extraction highlighted white, black, brown, yellow, and red as the most common descriptors. This suggests that the feather_color field, although text-based, still

contains meaningful recurring categories that could support future classification tasks after further normalization.

The sex field was strongly imbalanced, with Unknown accounting for the overwhelming majority of records. Because of this, the dataset is not well suited for sex-based comparisons. Yearly counts were also uneven, with a visible spike in 2023, indicating that temporal trends should be interpreted as descriptive rather than causal.

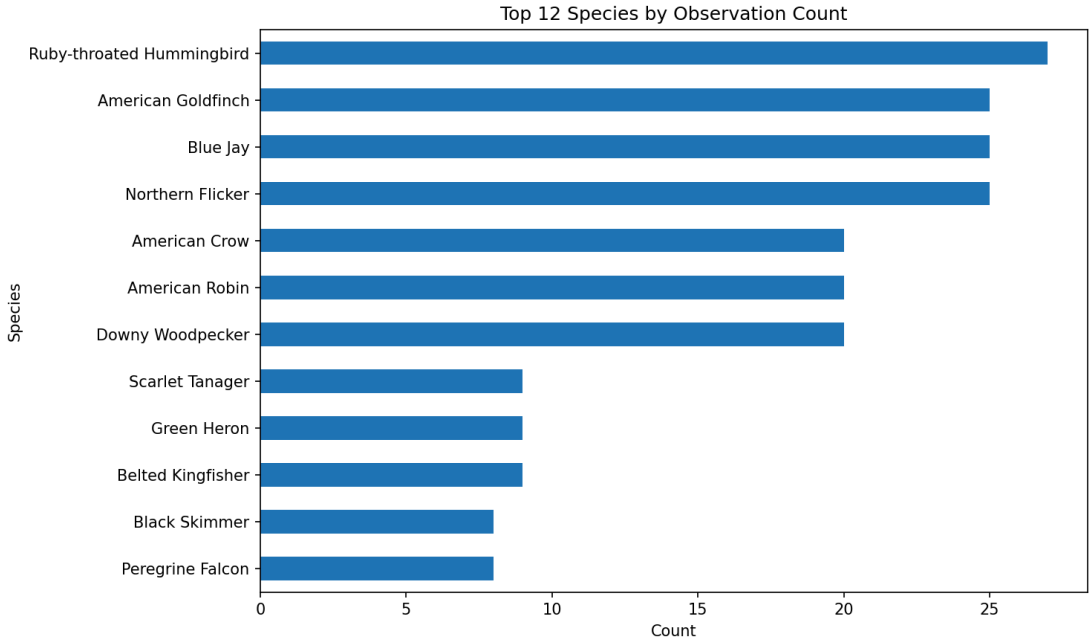

Figure 1. Top 12 Species by Observation Count.

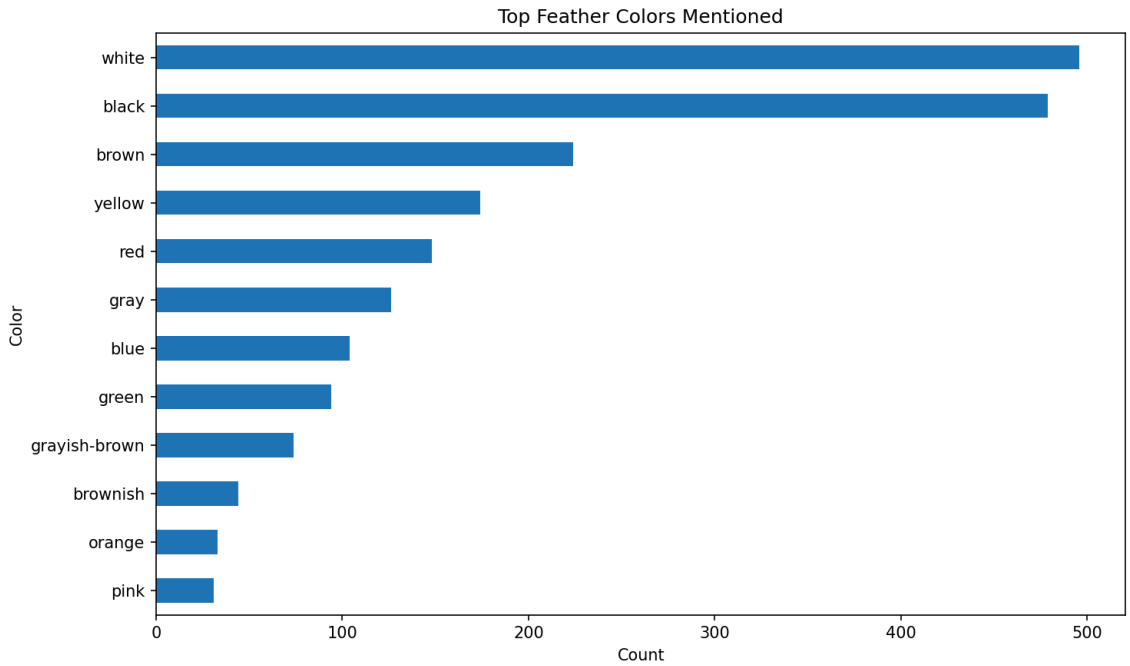

Figure 2. Top Feather Colors Mentioned.

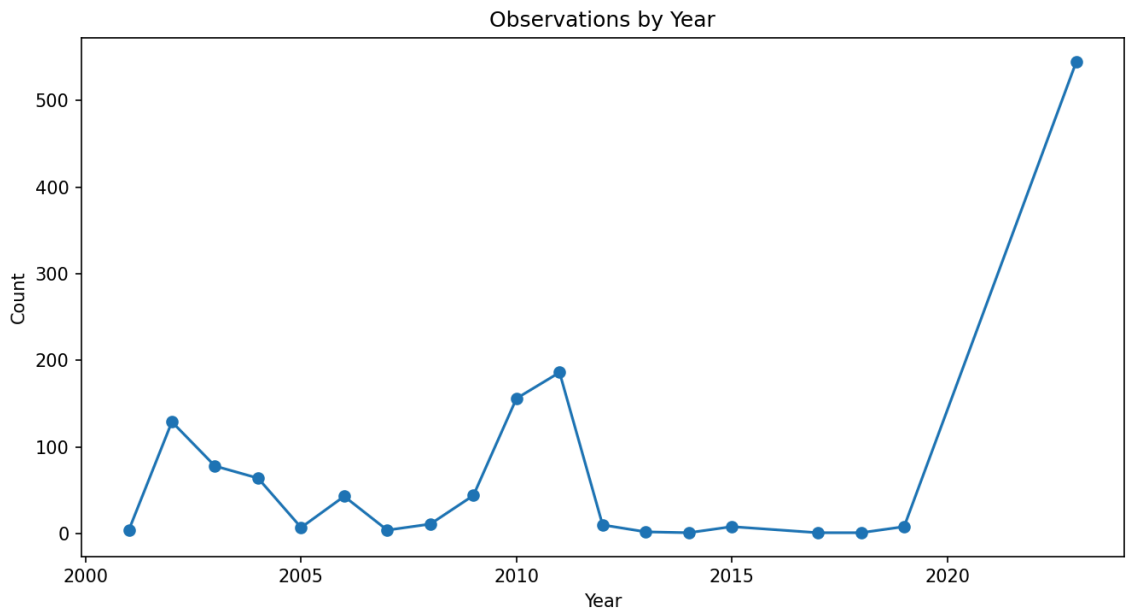

Figure 3. Observations by Year.

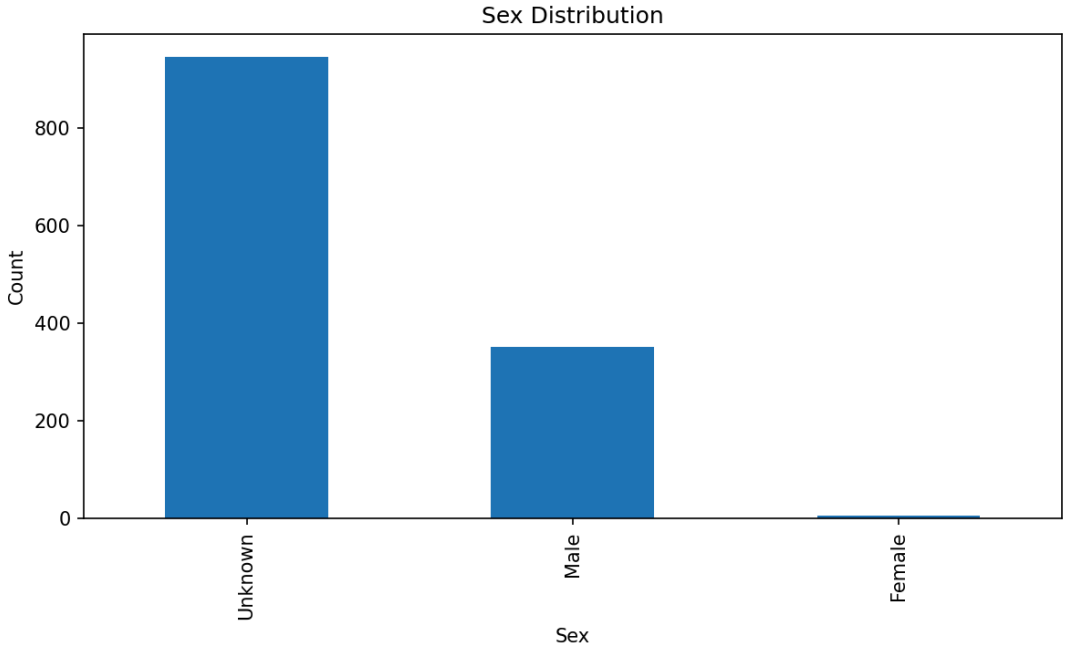

Figure 4. Sex Distribution.

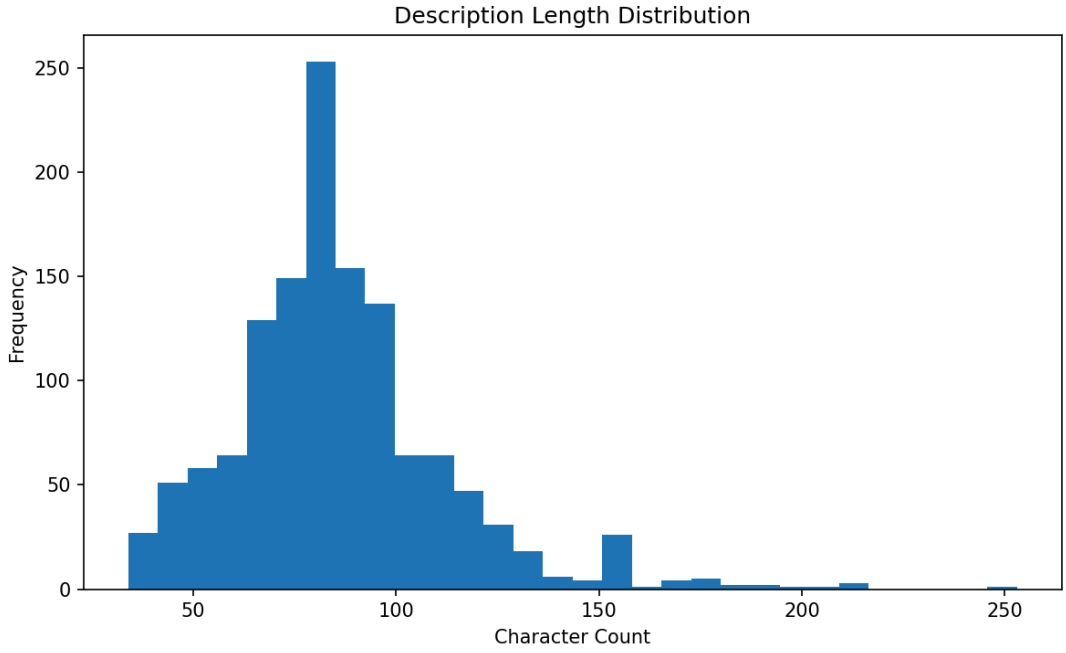

Figure 5. Description Length Distribution.

5 Discussion

The project demonstrates that a structurally unreliable CSV file can still be transformed into a useful analytical asset when the reconstruction strategy is guided by a stable anchor field. In this case, the timestamp column provided the structural reference needed to rebuild broken rows. The resulting analysis confirms that the dataset contains enough consistency for exploratory work, especially in species counts, recurring feather colors, and broad observation timing patterns.

At the same time, the analysis also shows why technical documentation is essential in data cleaning projects. The final visualizations are meaningful only because the report explains the assumptions used during reconstruction, the imbalance in important fields, and the risk of over-interpreting broad location labels or sparse temporal segments.

5.1 Limitations

1. The raw CSV file required reconstruction, so the final dataset depends on parsing assumptions.
2. The location field mixes broad and specific labels, reducing geographic precision.
3. The sex field is heavily dominated by Unknown values, limiting biological comparison.
4. Temporal coverage is uneven, so observed yearly spikes may reflect collection artifacts rather than real ecological shifts.
5. Feather-color labels remain text-based and may still contain partial inconsistencies.

6 Conclusion

This project successfully reconstructed, cleaned, and analyzed the unprocessed Birds of the World dataset using Python. By identifying the timestamp field as a structural anchor, malformed rows were rebuilt into a six-column dataset that could be cleaned, summarized,

and visualized. The analysis revealed broad species diversity, recurring feather-color patterns, an imbalanced sex field, and uneven observation counts across years.

Overall, the workflow supports the conclusion that damaged raw data can still yield useful descriptive insights when reconstruction is carefully documented. Future improvements could include richer text normalization, location standardization, and more advanced modeling once field consistency is improved.

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